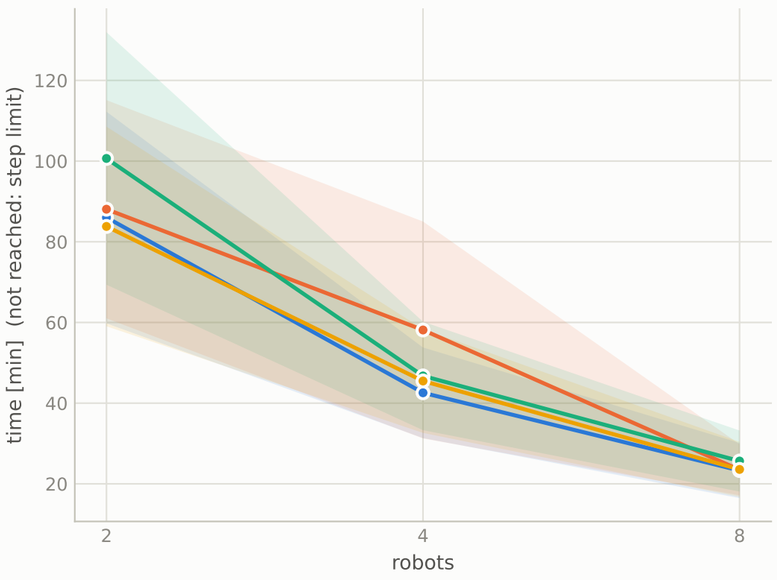
Multi-agent coverage: method comparison

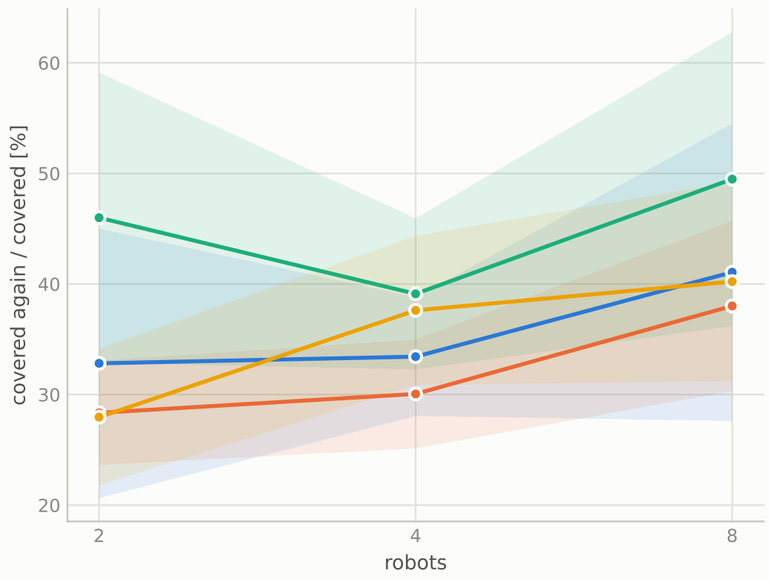
23 maps x 1 episodes per method and team size, deterministic actions, same start poses for every method. Lines: mean, bands: 95 % CI.

MAPPO · IPPO · MAPPO 5 m (comm. 5 m) · MAPPO tested at 5 m (comm. 5 m)

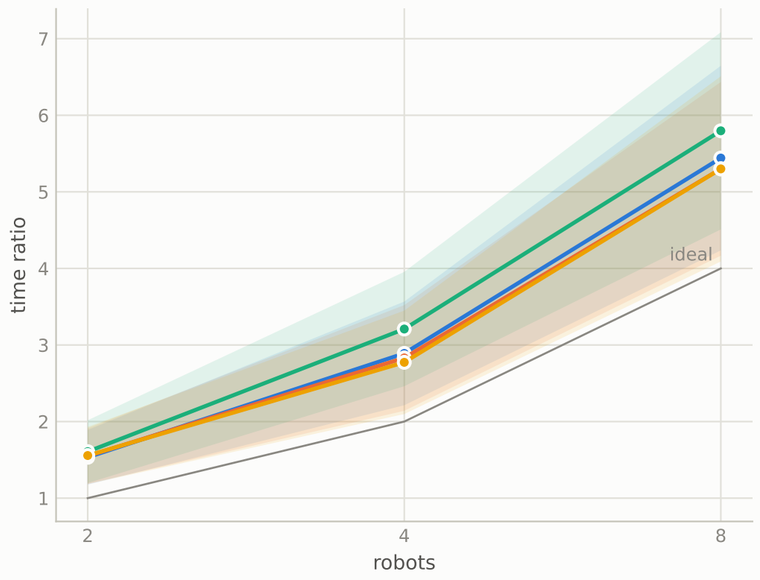
Time to 95 % coverage



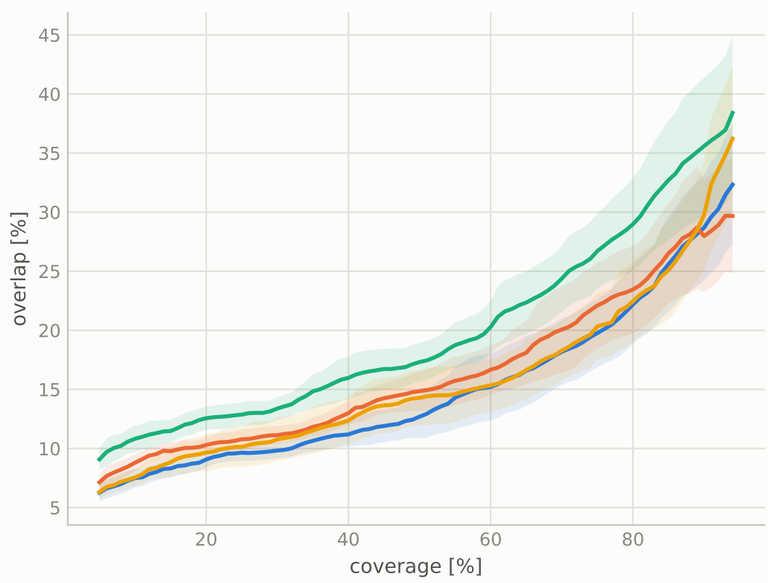
Overlap when 95 % coverage is reached



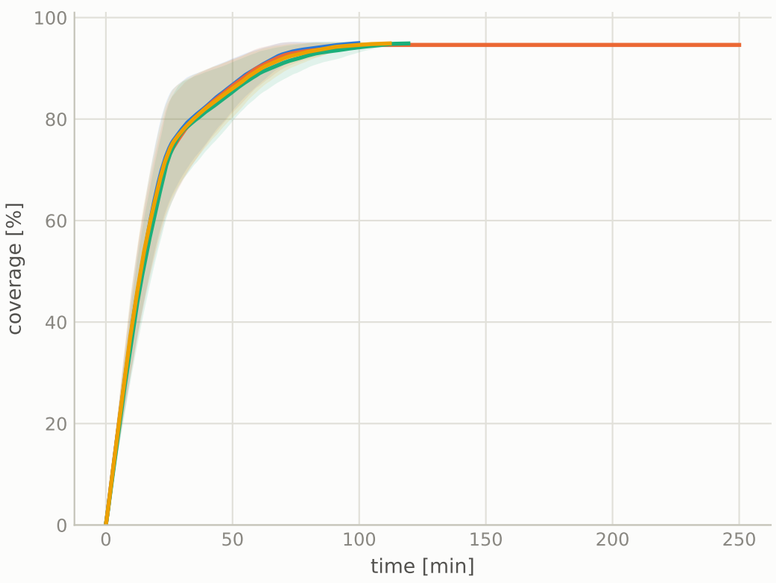
Speedup over 2 robots



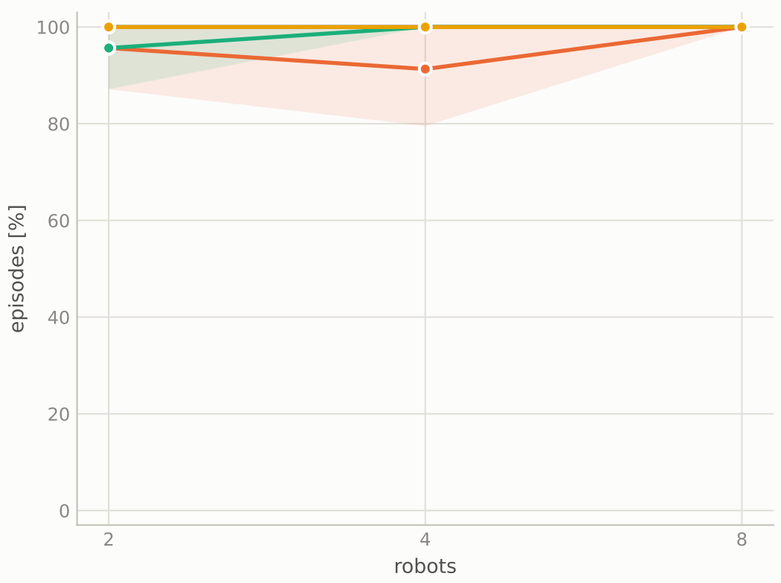
Overlap while covering, 4 robots



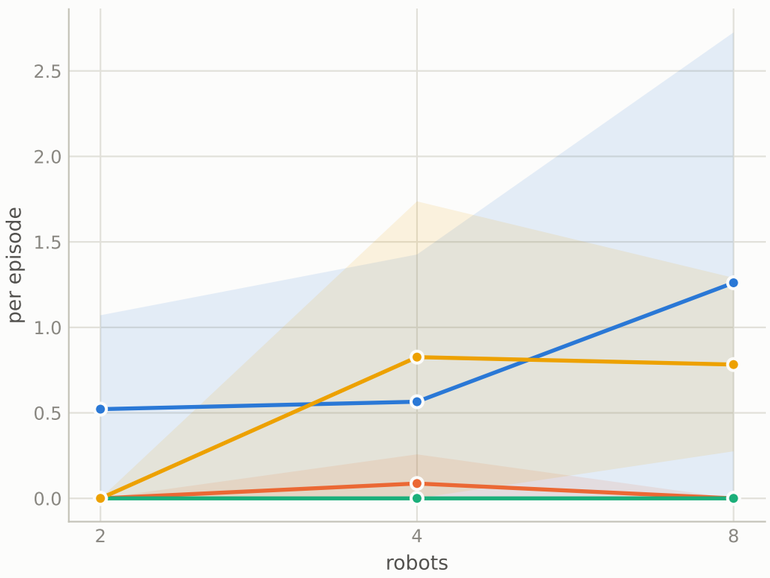
Coverage over time, 4 robots



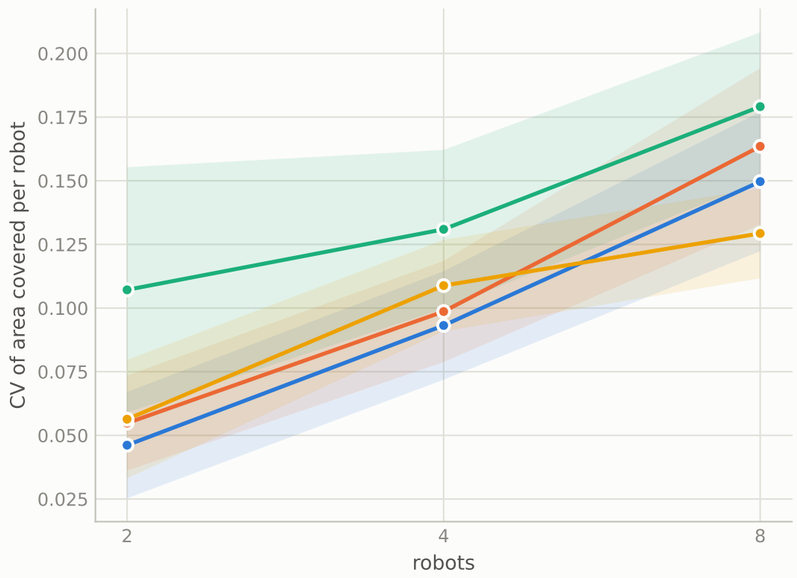
Success rate (95 % coverage)



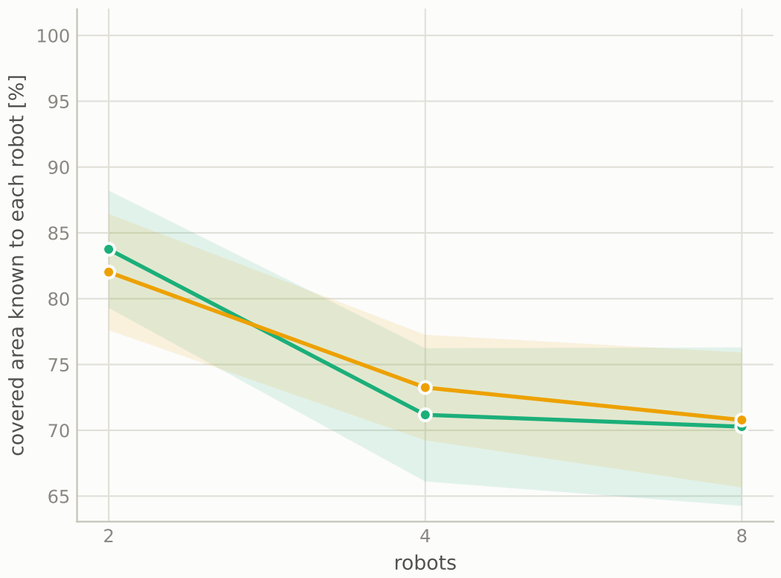
Collisions between robots



Workload imbalance



Belief recall (limited communication)



Summary, 4 robots (mean ± 95 % CI over episodes)

Method	Comm.	Success	Time to 90 %	Time to 95 %	Overlap at 95 %	Final overlap	Robot coll.	Imbalance	Belief recall	Connectivity
MAPPO	unlimited	100 %	38.9 ± 10.3 min	42.6 ± 11.2 min	33.4 ± 5.4 %	33.4 ± 5.4 %	0.6 ± 0.9	0.09 ± 0.02	100 ± 0 %	100 ± 0 %
IPPO	unlimited	91 %	38.2 ± 10.4 min	39.9 ± 11.6 min	30.1 ± 4.9 %	69.7 ± 54.3 %	0.1 ± 0.2	0.10 ± 0.02	100 ± 0 %	100 ± 0 %
MAPPO 5 m	5 m	100 %	42.8 ± 12.2 min	46.7 ± 13.4 min	39.1 ± 6.8 %	39.1 ± 6.8 %	0.0 ± 0.0	0.13 ± 0.03	71 ± 5 %	19 ± 4 %
MAPPO tested at 5 m	5 m	100 %	40.1 ± 11.3 min	45.5 ± 12.9 min	37.6 ± 6.7 %	37.6 ± 6.7 %	0.8 ± 0.9	0.11 ± 0.02	73 ± 4 %	21 ± 5 %